

	Test Cases – City Skyline: การหาช่วงตึกที่มองเห็นได้ด้วย Monotonic Stack กลุ่มที่ 8				
Test ID	ประเภทการทดสอบ	Input / Action	Expected Result	Actual Result	Status
TC01	Normal Case	[4, 5, 2, 10, 8]	Algorithm A = [5, 10, 10, -1, -1] Algorithm B = [5, 10, 10, -1, -1] Result comparison = PASS	Algorithm A: [5, 10, 10, -1, -1] Algorithm B: [5, 10, 10, -1, -1] Largest Rectangle: 16 Result comparison: PASS	PASS
TC02	Best Case / Increasing	[1, 2, 3, 4, 5]	Algorithm A = [2, 3, 4, 5, -1] Algorithm B = [2, 3, 4, 5, -1] Result comparison = PASS	Algorithm A = [2, 3, 4, 5, -1] Algorithm B = [2, 3, 4, 5, -1] Largest Rectangle: 9 Result comparison: PASS	PASS
TC03	Worst Case / Decreasing	[5, 4, 3, 2, 1]	Algorithm A = [-1, -1, -1, -1, -1] Algorithm B = [-1, -1, -1, -1, -1] Result comparison = PASS	Algorithm A: [-1, -1, -1, -1, -1] Algorithm B: [-1, -1, -1, -1, -1] Largest Rectangle: 9 Result comparison: PASS	PASS
TC04	Equal Height	[5, 5, 5, 5]	Algorithm A = [-1, -1, -1, -1] Algorithm B = [-1, -1, -1, -1] Result comparison = PASS	Algorithm A: [-1, -1, -1, -1] Algorithm B: [-1, -1, -1, -1] Largest Rectangle: 20 Result comparison: PASS	PASS
TC05	Alternating High-Low	[2, 8, 3, 7, 1, 6]	Algorithm A = [8, -1, 7, -1, 6, -1] Algorithm B = [8, -1, 7, -1, 6, -1] Result comparison = PASS	Algorithm A: [8, -1, 7, -1, 6, -1] Algorithm B: [8, -1, 7, -1, 6, -1] Largest Rectangle: 9 Result comparison: PASS	PASS
TC06	Boundary Case / Single Building	[10]	Algorithm A = [-1] Algorithm B = [-1] Result comparison = PASS	Algorithm A: [-1] Algorithm B: [-1] Largest Rectangle: 10 Result comparison: PASS	PASS
TC07	Target Not Found	[10, 8, 6, 4, 2]	Algorithm A = [-1, -1, -1, -1, -1] Algorithm B = [-1, -1, -1, -1, -1] Result comparison = PASS	Algorithm A: [-1, -1, -1, -1, -1] Algorithm B: [-1, -1, -1, -1, -1] Largest Rectangle: 18 Result comparison: PASS	PASS
TC08	Mandatory Case / Tallest at First	[10, 6, 8, 4, 2]	Algorithm A = [-1, 8, -1, -1, -1] Algorithm B = [-1, 8, -1, -1, -1] Result comparison = PASS	Algorithm A: [-1, 8, -1, -1, -1] Algorithm B: [-1, 8, -1, -1, -1] Largest Rectangle: 18 Result comparison: PASS	PASS
TC09	Mandatory Case / Tallest at Last	[2, 3, 4, 5, 10]	Algorithm A = [3, 4, 5, 10, -1] Algorithm B = [3, 4, 5, 10, -1] Result comparison = PASS	Algorithm A: [3, 4, 5, 10, -1] Algorithm B: [3, 4, 5, 10, -1] Largest Rectangle: 12 Result comparison: PASS	PASS
TC10	Empty Input / Invalid Boundary	n = 0	Program rejects the input and displays: Number of buildings must be greater than 0.	Program rejects the input and displays: Number of buildings must be greater than 0.	PASS
TC11	Invalid Input / Non-integer	Enter number of buildings: abc	Program rejects the input and displays: Invalid input. Please enter an integer.	Program rejects the input and displays: Invalid input. Please enter an integer.	PASS
TC12	Invalid Input / Negative Height	Height = -5	Program rejects the input and displays: Height must not be negative.	Program rejects the input and displays: Height must not be negative.	PASS
TC13	Boundary Case / Minimum Valid Size	[1, 2]	Algorithm A = [2, -1] Algorithm B = [2, -1] Result comparison = PASS	Algorithm A: [2, -1] Algorithm B: [2, -1] Largest Rectangle: 2 Result comparison: PASS	PASS
TC14	System-Specific / Largest Rectangle	[4, 5, 2, 10, 8]	Largest Rectangle = 16 NGE results from Algorithm A and B must match.	Algorithm A: [5, 10, 10, -1, -1] Algorithm B: [5, 10, 10, -1, -1] Largest Rectangle: 16 Result comparison: PASS	PASS

Test ID	ประเภทการทดสอบ	ภาพผลการรันจริง	หมายเหตุ
TC01	Normal Case	Input: [4, 5, 2, 10, 8] Algorithm A: Brute Force Result: [5, 10, 10, -1, -1] Time: 7900 ns Operations: 5 Algorithm B: Monotonic Stack Result: [5, 10, 10, -1, -1] Time: 1972100 ns Operations: 13 Largest Rectangle: 16 Result comparison: PASS	ทดสอบกรณีปกติ และตรวจสอบว่า Algorithm A และ B ให้ผลลัพธ์ตรงกัน
TC02	Best Case / Increasing	Input: [1, 2, 3, 4, 5] Algorithm A: Brute Force Result: [2, 3, 4, 5, -1] Time: 16400 ns Operations: 4 Algorithm B: Monotonic Stack Result: [2, 3, 4, 5, -1] Time: 2978000 ns Operations: 13 Largest Rectangle: 9 Result comparison: PASS	ทดสอบกรณีอาคารสูงขึ้นต่อเนื่อง ซึ่งเป็น Best Case ของ Algorithm A
TC03	Worst Case / Decreasing	Input: [5, 4, 3, 2, 1] Algorithm A: Brute Force Result: [-1, -1, -1, -1, -1] Time: 11500 ns Operations: 10 Algorithm B: Monotonic Stack Result: [-1, -1, -1, -1, -1] Time: 1223700 ns Operations: 9 Largest Rectangle: 9 Result comparison: PASS	ทดสอบกรณีอาคารเตี้ยลงต่อเนื่อง ซึ่งเป็น Worst Case ของ Algorithm A
TC04	Equal Height	Input: [5, 5, 5, 5] Algorithm A: Brute Force Result: [-1, -1, -1, -1] Time: 15300 ns Operations: 6 Algorithm B: Monotonic Stack Result: [-1, -1, -1, -1] Time: 978100 ns Operations: 7 Largest Rectangle: 20 Result comparison: PASS	ทดสอบกรณีอาคารมีความสูงเท่ากันทั้งหมด และค่าที่เท่ากันไม่ถือเป็น Next Greater Element
TC05	Alternating High-Low	Input: [2, 8, 3, 7, 1, 6] Algorithm A: Brute Force Result: [8, -1, 7, -1, 6, -1] Time: 7400 ns Operations: 9 Algorithm B: Monotonic Stack Result: [8, -1, 7, -1, 6, -1] Time: 718200 ns Operations: 16 Largest Rectangle: 9 Result comparison: PASS	ทดสอบกรณีความสูงของอาคารสลับสูง - ต่ำ

Test ID	ประเภทการทดสอบ	ภาพผลการรันจริง	หมายเหตุ
TC06	Boundary Case / Single Building	<pre>Input: [10] Algorithm A: Brute Force Result: [-1] Time: 25100 ns Operations: 0 Algorithm B: Monotonic Stack Result: [-1] Time: 678200 ns Operations: 1 Largest Rectangle: 10 Result comparison: PASS</pre>	ทดสอบกรณีมีอาคารเพียงหนึ่งหลัง ซึ่งควรได้ -1
TC07	Target Not Found	<pre>Input: [10, 8, 6, 4, 2] Algorithm A: Brute Force Result: [-1, -1, -1, -1, -1] Time: 13099 ns Operations: 10 Algorithm B: Monotonic Stack Result: [-1, -1, -1, -1, -1] Time: 2755000 ns Operations: 9 Largest Rectangle: 18 Result comparison: PASS</pre>	ทดสอบกรณีไม่มีอาคารที่สูงกว่าทางด้านขวา
TC08	Mandatory Case / Tallest at First	<pre>Input: [10, 6, 8, 4, 2] Algorithm A: Brute Force Result: [-1, 8, -1, -1, -1] Time: 32601 ns Operations: 8 Algorithm B: Monotonic Stack Result: [-1, 8, -1, -1, -1] Time: 434700 ns Operations: 11 Largest Rectangle: 18 Result comparison: PASS</pre>	ทดสอบกรณีอาคารที่สูงที่สุดอยู่ตำแหน่งแรก
TC09	Mandatory Case / Tallest at Last	<pre>Input: [2, 3, 4, 5, 10] Algorithm A: Brute Force Result: [3, 4, 5, 10, -1] Time: 12200 ns Operations: 4 Algorithm B: Monotonic Stack Result: [3, 4, 5, 10, -1] Time: 521500 ns Operations: 13 Largest Rectangle: 12 Result comparison: PASS</pre>	ทดสอบกรณีอาคารที่สูงที่สุดอยู่ตำแหน่งสุดท้าย
TC10	Empty Input / Invalid Boundary	<pre>Enter number of buildings: 0 Number of buildings must be greater than 0.</pre>	ทดสอบกรณีข้อมูลว่างหรือจำนวนอาคารเป็น 0 และการตรวจสอบ Input
TC11	Invalid Input / Non-integer	<pre>Enter number of buildings: a b c Invalid input. Please enter an integer.</pre>	ทดสอบการป้อนข้อมูลที่ไม่ใช่จำนวนเต็ม

Test ID	ประเภทการทดสอบ	ภาพผลการรันจริง	หมายเหตุ
TC12	Invalid Input / Negative Height	<pre>Enter number of buildings: 1 Enter height of building 1: -5 Height must not be negative.</pre>	ทดสอบการป้อนค่าความสูงติดลบ
TC13	Boundary Case / Minimum Valid Size	<pre>Input: [1, 2] Algorithm A: Brute Force Result: [2, -1] Time: 9500 ns Operations: 1 Algorithm B: Monotonic Stack Result: [2, -1] Time: 537400 ns Operations: 4 Largest Rectangle: 2 Result comparison: PASS</pre>	ทดสอบ Boundary Case ที่มีจำนวนอาคารขั้นต่ำที่โปรแกรมยอมรับ
TC14	System-Specific / Largest Rectangle	<pre>Input: [4, 5, 2, 10, 8] Algorithm A: Brute Force Result: [5, 10, 10, -1, -1] Time: 10800 ns Operations: 5 Algorithm B: Monotonic Stack Result: [5, 10, 10, -1, -1] Time: 419500 ns Operations: 13 Largest Rectangle: 16 Result comparison: PASS</pre>	ทดสอบความถูกต้องของ Largest Rectangle และตรวจสอบว่า Algorithm A และ B ให้ผล NGE ตรงกัน